



CYT_THEORY ASSIGNMENT - 2

K V S CHANAKYA VAMSI | 261100690013 | 22.08.2026

Compare Permissions on Regular Files & Directories

Permission Testing Table:

Permission	Regular File (sample.txt) - Test	Observation	Directory (sample_dir) - Test	Observation
400 (r--)	Read and modify	Read: Yes. Modify: No	List and enter	List: No. Enter: No.
200 (-w-)	Read and modify	Read: No. Modify: Yes.	List, enter, create	List: No. Enter: No. Create: No, because x is absent.
100 (--x)	Read and execute	Read: No. Execute: Yes.	List and enter	List: No. Enter: Yes.
500 (r-x)	Read and modify	Read: Yes. Modify: No. Execute: Yes.	List and enter	List: Yes. Enter: Yes.
600 (rw-)	Read and modify	Read: Yes. Modify: Yes.	List, enter, create	List: No. Enter: No. Create: No.
700 (rwx)	Read, modify, execute	Read: Yes. Modify: Yes. Execute: Yes.	List, enter, create, delete	List: Yes. Enter: Yes. Create/Delete: Yes.

Questions and Answers

1. What does read (r) permission allow you to do with a regular file?

It allows you to **view/read the contents** of the file.

2. What does read (r) permission allow you to do with a directory?

It allows you to **list the names of files and directories** contained within it.

3. What does write (w) permission mean for a regular file?

It allows you to **modify or change the contents** of the file.

4. What does write (w) permission mean for a directory?

It allows you to **create, delete, and rename directory entries**, subject to the directory's other permissions and any special restrictions.

5. What does execute (x) permission mean for a regular file?

It allows the file to be **executed as a program or script**, provided the file is actually executable in a meaningful way.

6. Why is execute (x) permission important for accessing a directory?

For a directory, x means **search/traverse permission**. It allows you to enter the directory with `cd` and access items within it when the relevant permissions are also satisfied.

7. Which permission combination allows the owner to fully access a regular file?

700 (rwx) — read, write, and execute.

8. Which permission combination allows the owner to fully access a directory?

700 (rwx) — list contents, enter/traverse, and create/delete/rename entries.

9. Based on your observations, explain why the same r, w, and x permissions behave differently for a regular file and a directory.

The meaning of permissions depends on the type of filesystem object. For a **regular file**, r controls reading contents, w controls modifying contents, and x controls execution. For a **directory**, r controls listing its entries, w controls creating/deleting/rename entries, and x controls entering or traversing the directory. Therefore, the same permission letters provide different capabilities depending on whether the object is a file or directory.